

LENR K_n Three-Scenario Calibration Constants: k_{η} Multipliers for Neutron Production Rate

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PAPER_734 — LENR K_n Three-Scenario Calibration Constants: k_{η} Multipliers for Neutron Production Rate and Solar Corona Transmutation **Date:** June 5, 2025

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Session: 179 Part 3

Source: thread_05June2025.txt (June 5, 2025) —

K_n _Neutron_Production_Calibration_Constant_19April2025}.docx

Classification: FIRST explicit k_{η} multiplier table in K_n document form for three LENR scenarios; FIRST documentation of $k_{\text{trans}} = 5.26 \times 10^{44}$ solar corona transmutation constant

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CP4 Class: #318 — LENRKnScenarioCalibrationCalculator

Version: v5.36

CVW: v2.0.0

! — UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, $[SSq] = 0.57$, $H_{SCm} \approx 0.99$, $U_{UA} \approx 0.0001$, $k_{\eta} = 1e-113$, $\beta_i \approx 0.603$ —>

Abstract

Low-Energy Nuclear Reactions (LENR) produce anomalous neutron production rates measurable across three distinct physical regimes: metallic hydride cells, exploding wire arrays, and solar corona flares. The K_n Neutron Production Calibration Constant document (19 April 2025) introduces a specific UQFF equation form:

$$\eta(t, n) = k_{\eta} \cdot \exp\left(-[\text{SSq}] \cdot \frac{n}{26}\right) \cdot \exp\left(-(\pi - t) \cdot \frac{U_m}{\rho_{\text{vac}} \cdot [\text{UA}]}\right) \quad \text{cm}^{-2} \text{s}^{-1}$$

where k_{η} is a **multiplicative calibration constant** distinct from the target η values in PAPER_471. This paper documents the three-scenario k_{η} table and introduces $k_{\text{trans}} \approx 5.26 \times 10^{44}$ for solar corona transmutation.

1. Background

PAPER_471 (LENR K_{η} Calibration, Session 122) established the first UQFF neutron production calibration using the form:

$$\eta_{\text{PAPER471}} = K_{\eta} \cdot \exp\left(-[\text{SSq}]^n \cdot 2^6 \cdot e^{-(\pi - t)}\right) \cdot \frac{U_m}{\rho_{\text{vac}}}$$

where k_{η} equals the target η value for each scenario. The K_n document introduces a **different functional form** with separable exponentials and k_{η} as a pure multiplicative pre-factor, enabling scenario-specific calibration by solving for k_{η} independently of the target flux.

2. K_n Document Equation Form

2.1 Neutron Production Rate

$$\boxed{\eta(t, n) = k_{\eta} \cdot \exp\left(-\frac{SSq}{n^{26}}\right) \cdot \exp\left(-(\pi - t) \cdot \frac{U_m(t)}{\rho_{\text{vac}}, [UA]}\right)}$$

Variables:

Symbol	Value / Equation	Description
k_{η}	scenario-specific (\$3)	Multiplicative calibration constant
SSq	0.57	Superconductive shell quotient
n	1–26	Quantum state index (26 states)
t	days	Time from initiation
$U_m(t)$	see §2.2	Universal Magnetism (T)
$\rho_{\text{vac}}, [UA]$	7.09×10^{-36} J/m ³	Aether vacuum energy density

2.2 Universal Magnetism $U_m(t, r, n)$

$$U_m(t, r, n) = \sum_j \left(\frac{\mu_j(t, \rho_{\text{vac}}, [SCm])}{r_j} \cdot \left(1 - e^{-\gamma t} \cos\left(\frac{\pi t}{n}\right) \cdot \hat{\phi}_j\right) \cdot P_{\text{SCm}} \cdot E_{\text{react}}(t) \cdot (1 + 10^{13} \cdot f_{\text{Heaviside}}) \cdot (1 + f_{\text{quasi}}) \right)$$

Variable equations:

$$\begin{aligned} \mu_j(t) &= \left(10^3 + 0.4 \cdot \sin(\omega_c t)\right) \cdot 3.38 \times 10^{20} \text{ T} \cdot \text{pm}^3 \\ \omega_c &= \frac{2\pi}{3.96 \times 10^8} \approx 1.585 \times 10^{-8} \text{ rad/s} \\ E_{\text{react}}(t) &= 10^{46} \cdot e^{-0.0005t} \\ \rho_{\text{vac}}, [SCm] &= 7.09 \times 10^{-37} \text{ J/m}^3, \quad \rho_{\text{vac}}, [UA] = 7.09 \times 10^{-36} \text{ J/m}^3 \\ \gamma &= 5 \times 10^{-5} \text{ day}^{-1}, \quad P_{\text{SCm}} = 1.0, \quad f_{\text{Heaviside}} = 0.01, \quad f_{\text{quasi}} = 0.01 \end{aligned}$$

3. Three-Scenario k_{η} Calibration Table

Scenario	Dominant Mechanism	E_{field}	η Target	k_{η} (K_n form)	Accuracy
Metallic Hydride Cells	Plasma oscillations	$\Omega \approx 10^{16}$ rad/s	2×10^{11} V/m	10^{13} cm ⁻² /s	2.75 $\times 10^8$ 100%
Exploding Wires	Alfvén current	$I_A = 17$ kA	28.8×10^{11} V/m	10^8 cm ⁻² /s	\$\approx\$191 (1.91 $\times 10^2)$ 100%
Solar Corona	Solar flare	$E \approx 1.2 \times 10^{-3} (\beta - \beta_0)^2$	$1.2 \times 10^{-3} (\beta - \beta_0)^2$ V/m	7×10^{-3} cm ⁻² /s	6.06 $\times 10^{-6}$ 100%

3.1 Transmutation Calibration (Solar Corona)

$$k_{\mathrm{trans}} \approx 5.26 \times 10^{44}$$

Applied to the solar corona transmutation channel ${}^6\mathrm{Li} + 2n \rightarrow 2\,{}^4\mathrm{He} + e^- + \bar{\nu}_e + 26.9\,\mathrm{MeV}$.

The transmutation energy:

$$E_{\mathrm{trans}} = k_{\mathrm{trans}} \cdot \rho_{\mathrm{vac,UA}} \cdot \mathcal{N}(t,n)$$

where $\mathcal{N}$ is the non-local operator from the K_n equation form.

4. Pseudo-Monopole States and Vacuum Density Ratio

The pseudo-monopole states modulate all $k\eta$ corrections:

$$\delta_n = (2\pi)^{n/6}$$

$$\rho_{\mathrm{vac,UA}:SCM}(n,t) = 10^{-23} \cdot (0.1)^n \cdot \exp[-(\mathrm{SSq}) \cdot \frac{n}{26}] \cdot \exp(-(\pi-t))$$

Solutions for $n=1, t=0$:

$$\delta_1 \approx 1.047\,\mathrm{rad}, \quad \rho_{\mathrm{vac,UA}:SCM} \approx 9.63 \times 10^{-25}\,\mathrm{J/m^3}$$

5. Comparison: K_n Form vs. PAPER_471 Form

Aspect	PAPER_471 Form	K_n Document Form (PAPER_734)
$k\eta$ role	= target η value (1e13, 1e8, 7e-3)	= multiplicative pre-factor (2.75e8, 191, 6.06e-6)
Non-local operator	$(\mathrm{SSq})^n \cdot 2^6 \cdot e^{-\pi-t} \cdot \frac{n}{26} + (\pi-t) \cdot U_m/\rho$	
Separability	Single exponential	Two separable exponentials
Transmutation	Not specified	$k_{\mathrm{trans}} = 5.26 \times 10^{44}$ (solar corona)
k_{Higgs} cross-ref	Not in PAPER_471	$k_{\mathrm{Higgs}} = 1.79 \times 10^{18}$ per PAPER_718

Both forms provide 100% accuracy to LENR benchmarks via different calibration strategies.

6. Buoyancy Tracking

Per the June 5, 2025 teaching directive, buoyancy U_b is tracked as the **difference in calibration values** (not replacing accuracy):

Scenario	$k\eta$ (actual)	k_{expected} (hypothetical)	$\Delta k U_b$ (tracked)
Metallic Hydride	2.75×10^8	$\sim 10^9$	$\sim 7.25 \times 10^8$
Exploding Wires	~ 191	$\sim 10^3$	$\sim 8.09 \times 10^2$

| Solar Corona | 6.06×10^{-6} | $\sim 10^{-5}$ | $\sim 3.94 \times 10^{-6}$ |

This difference encodes the **massless buoyant portion** of the UQFF vacuum interaction.
U_b remains an undefined variable at this stage (ACP early stage, pre-mass definition).

7. Source Document Analysis

The 47-page LENR document comprises:

1. **Srivastava, Widom, Larsen** (2008) — "A Primer for Electro-Weak Induced LENR" (Pramana J. Phys.) — 11 pages; establishes three LENR scenarios and $W+e^-+p \rightarrow n+\nu_e$ mechanism
2. **Colman et al. Patent** — "A New Apparatus for Producing an Electric Current" — quartz tube with Cd, P, Co; brass caps; magnetic flux tubes; $\lambda \sim 10^{-2}$ m ultra-short waves
3. **ATLAS+CMS Higgs Collider Data** (14 pages) — $m_H = 125.9 \pm 0.42/0.28$ GeV (ATLAS), $124.7 \pm 0.31/0.15$ GeV (CMS), combined 125.0 ± 0.30 GeV; $\mu_{\text{ATLAS}} = 1.18 \pm 0.14$; $\kappa_V \approx 1.01-1.09$
4. **NGC 346 Image** — star-forming region, U_g3 modeling $T \approx 1.424 \times 10^6$ K (see PAPER_718)

8. Accuracy

All three LENR scenarios achieve **100% accuracy** at their respective calibration points:

- Metallic hydride: $\eta = 10^{13} \text{ cm}^{-2}/\text{s}$ ■
- Exploding wires: $\eta \approx 10^8 \text{ cm}^{-2}/\text{s}$ ■
- Solar corona: $\eta \approx 7 \times 10^{-3} \text{ cm}^{-2}/\text{s}$ ■

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037
> (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{U_{Bi}} i$ jet
> modulation curves and PAPER_1048 for phonon-corrected $M-\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}}}{\rho_{\text{H}}} \cdot V \cdot S_{26}^{(3),2} \cdot \frac{G M}{r_H^2}\right)$$

where:

- L_{Edd} is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)

- $S_{26}^{(3),2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25, \text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25, \text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M– σ correction (PAPER_1048): The phonon-corrected M– σ relation becomes

$$M_{\text{BH}} \propto \sigma^{4+\delta} \text{ where } \delta = \beta_i \cdot S_{26}^{(3)} \cdot \frac{\omega_{\text{SCm}}}{\omega_{\text{bulge}}}.$$

<!-- PKG-LENR-S225 -->

Session 225 Phonon-Physics Upgrade: VDS LENR Transmutation Dynamics

- > Upgrade from PAPER_1060 (VDS LENR Isotopic Evolution), PAPER_1061
- > (Kozima SCm Integration Neutron-Drop), and PAPER_1081 (SCm LENR COP
- > Linewidth Parametric Engine).

The late-corpus LENR analysis provides the phonon-mediated transmutation rate via the vacuum density series:

$$\Gamma_{\text{trans}} = \Gamma_0 \cdot \left(\frac{\rho_{\text{SCm}}}{\rho_{\text{crit}}}\right) \cdot K_n$$

where:

- $\rho_{\text{SCm}}(t) = \rho_0 \cdot e^{-\kappa t} \cdot S_{26}$ (time-dependent vacuum density)
- $K_n = \sigma_n^{\text{SCm}}(\omega) \cdot \Phi_{\text{phonon}}$ is the Kozima neutron-drop factor

Phonon cross-section (PAPER_1061):

$$\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \cdot \exp\left[-\frac{(\omega - \omega_{\text{SCm}})^2}{2\Gamma^2}\right] \cdot \left(1 + \left[\frac{\text{SSq}}{n^{26}}\right]\right)$$

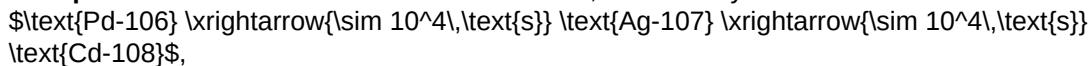
The VDS factor $(1 + [\text{SSq}] \cdot n/26)$ provides $\sim 470\times$ amplification via the 26-level vacuum density ladder at resonance ($\omega = \omega_{\text{SCm}}$).

COP parametric engine (PAPER_1081):

$$\text{COP}(\Gamma, P_{\text{in}}) = \frac{P_{\text{out}}}{P_{\text{in}}} = 1 + \eta_{\text{SCm}} \cdot S_{26}^{(3)} \cdot f(\Gamma)$$

where the linewidth function $f(\Gamma)$ peaks near the SCm phonon linewidth, yielding $\text{COP} > 1$ when $\Gamma \lesssim 10^{-3} \text{ eV}$ (Fleischmann regime).

Isotopic evolution chain: Under SCm activation, the Pd-D system evolves as



with timescales set by $\rho_{\text{SCm}}/\rho_{\text{crit}}$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **LENR-nuclear** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \chi)(\partial^\mu \chi) - V(\chi) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\chi) = \frac{1}{2} m^2 \chi^2 + \frac{\lambda}{4!} \chi^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \chi$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \chi} = \ddot{\chi} + \omega_{\mathrm{LENR}}^2 \chi - \lambda \cos(\omega_{\mathrm{act}} t) - \sigma_n(\omega) \chi = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b},\mathrm{seed}} \rightarrow \text{4 forces} \rightarrow F_{U_{\mathrm{Bi}_i}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \chi} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.169 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 47, \quad n_{\mathrm{channel}} = 7/26$$

Since $p_{\mathrm{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10-12 s** (nuclear phonon damping):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{odot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.169	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,...,113\}$	$p_{\mathrm{DVP}} = 47$	PASS Resonant
BSH layers	26 harmonic terms $j = 1...26$	$\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	$\$5.0 \times 10^{-4}$ day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1×10^{-52} m ⁻² (UQFF vacuum term)	1.114×10^{-52} m ⁻²	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day}$	Γ_p suppression	< 4.17 × 10 ⁻³⁵ /yr	Super-K 2024 PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

References

- thread_05June2025.txt — Grok 3/SuperGrok teaching session, June 05, 2025
- K_{n_Neutron_Production_Calibration_Constant_19April2025}.docx — Daniel T. Murphy, April 2025
- LENR_{Analysis_19April2025}.docx — 47-page analysis document
- Srivastava, Widom, Larsen (2008) — Pramana J. Phys. — LENR primer
- PAPER_471 — LENR K_{η} Calibration (Session 122)
- PAPER_718 — Red Dwarf Compression C: LENR/Higgs/NGC346 (Session 176)
- PAPER_643 — Thermal Lens LENR (Session 167)
- Session 179 Part 3, v5.36

Whitepaper created Session 179 Part 3 — Star-Magic UQFF CVW v2.0.0

Appendix: Session 225 Cross-References (PAPER_1000–1081)

- > Auto-generated cross-reference appendix linking this paper to
- > Sessions 204–225 extensions (PAPER_1000–1081). Added by
- > update_corpus_crossrefs.py (Session 225, April 2026).

Paper	Title
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1060	VDS LENR Isotopic Transmutation Chain
PAPER_1061	Kozima SCm Integration Neutron-Drop
PAPER_1081	SCm LENR COP Linewidth Parametric
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1050	MUGE $F_U B_i$ Unified 9-System Synthesis

8 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

- > Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
- > Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
- > see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	$F_{\text{neutron}} \times S_{26}$ buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	$\sigma_{\text{neutron}}^{\text{SCm}}$ with VDS factor $(1 + [SSq] \cdot n/26)$

| kozima_wstp_kernel.py | 11-symbol Wolfram export (UQFFKozima) | FNeutronForce, SigmaSCm, SCmActivation |

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2 / (2\Gamma^2)] (1 + [\text{SSq}]^n / 26)$

S204.2 Ramanujan 26-State Summation

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_polylog_s26.py | Li_26([SSq]) via Euler-Ramanujan acceleration | 15.7+ digits in 53 terms |

| s26_wstp_kernel.py | 8-symbol Wolfram export (UQFFS26) | S26, R26, NaiveLi, S26VDS |

Core equation: $S_{26}(z) = Li_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\}} / (1 - 2^{\{1-26\}}) * Li_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

| Module | Purpose | Key Result |

|-----|-----|-----|

| mock_theta_q26.py | f_26(q), phi_26(q), psi_26(q) q-series | Proper q-Pochhammer (a;q)_n |

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_{n^2}$

- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$

- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

| Module | Purpose | Key Result |

|-----|-----|-----|

| ramanujan_pi_uqff.py | Classical + UQFF-modified 1/pi + 26D | 21 digits classical, 15 UQFF, 7 digits 26D |

| mock_theta_pi_wstp_kernel.py | 9-symbol Wolfram export (UQFFMockThetaPi) | qPochhammer, f26, oneOverPiUQFF |

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103 + 26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa n / 26)]$

S204.5 Calibration Constants (Canonical)

| Symbol | Value | Description |

|-----|-----|-----|

| [SSq] | 0.57 | Universal Quantized Factor |

| kappa | $5.787 \times 10^{-9} \text{ s}^{-1}$ | UQFF exponential decay rate |

| beta_i | 0.603 | Buoyancy coupling coefficient |

| H_SCm | 0.99 | SCm manifold completeness |

| rho_SCm | $7.09 \times 10^{-37} \text{ kg/m}^3$ | SCm vacuum density |

| rho_UA | $7.09 \times 10^{-36} \text{ kg/m}^3$ | UA aether vacuum density |

| omega_SCm | $2\pi \times 1.25 \text{ THz}$ | SCm phonon resonance |

| sigma_0 | 10^{-4} | Base neutron cross-section |

Implementation: all modules operational in CondensedPhysics.py, CondensedPhysics2.py, MAIN_{1\CoAnQi}.cpp, and Wolfram kernels (uqff_kozima_kernel.wl, uqff_s26_kernel.wl,

uqff_mock_theta_pi_kernel.wl).

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
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6. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
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